Problem Set 4

Student handout

Distributions, tails, and coupon collection

100 points + 10 optional challenge points • About 4 hours

Name: _____ **Due:** _____

Instructions. Show your reasoning and state each distribution's support and parameters. Geometric variables count trials including the first success; exponential parameters are rates. Write $H_r = \sum_{j=1}^{r} 1/j$ and $H_0 = 0$. Exact expressions suffice unless decimals are requested. A calculator or short program is allowed for numerical comparisons. Independent uniform choices of coupon types are assumed except in the challenge.

I. Discrete mechanisms

1. A count is not a wait [12 points]

A box contains 20 distinguishable gadgets, of which 6 are defective.

(a) Choose one gadget uniformly. Let I indicate a defect. Give its law, mean, and variance. (2)

(b) Make four independent uniform draws *with replacement*. For the number B of defective gadgets drawn, give the law, $\mathbb{P}(B = 2)$, mean, and variance. (3)

(c) Choose four gadgets *without replacement*. For the defective count D, give its support, PMF, mean, and variance. Explain the variance comparison with (b). (4)

(d) Draw independently with replacement until the first defect. For the draw count G, give its PMF, $\mathbb{P}(G > 4)$, mean, and variance. Why is it not binomial? (3)

2. Rare errors and a geometric tail [10 points]

Each of 2000 independent transmitted bits is corrupted with probability 0.001. Let N count corrupted bits.

(a) Give the exact law of N and a suitable Poisson approximation. Compare the exact and approximate $\mathbb{P}(N \geq 2)$ to four decimals. (4)

(b) For $Q \sim \text{Pois}(\lambda)$, derive $\mathbb{E}[Q(Q-1)] = \lambda^2$ by shifting the PMF sum, and recover its variance. (3)

(c) In an unlimited stream with the same independent error probability, let G be the bit index of the first error. Use $\mathbb{E}[G] = \sum_{m=0}^{\infty} \mathbb{P}(G > m)$ to find its mean, and calculate $\mathbb{P}(G > s + t \mid G > s)$ for integers $s, t \geq 0$. (3)

II. What the coupon-collector mean leaves out

3. The last few stickers [14 points]

An album has n equally likely sticker types. Let $T_{n,r}$ be the number of purchases needed to see r distinct types, $1 \le r \le n$, and let $T_n = T_{n,n}$.

(a) Define stages W_k, for $0 \le k < n$, identify their laws, and justify their mutual independence from the conditional distribution after any completed history. (4)

(b) Derive $\mathbb{E}[T_{n,r}]$ and $\mathrm{Var}(T_{n,r})$ as finite sums. Simplify the expectation with harmonic numbers. (4)

(c) For $n = 100$, compare the expected purchases for the first 90 types with the expected *additional* purchases for the last 10. Give one decimal place. Why do the last 10 take so long? (3)

(d) For $n = 6$, find $\mathbb{E}[T_6]$ and $\mathrm{Var}(T_6)$ exactly. If exactly two faces are still missing, what is the expected remaining number of rolls? Why is T_6 not geometric? (3)

4. A deadline is not an expectation [14 points]

Let M_m count types missing after $m \ge 1$ purchases, with $n \ge 2$ types. Write $J_i = 1$ if type i is missing, $q = (1 - 1/n)^m$, and $b = (1 - 2/n)^m$.

(a) Compute $\mathbb{E}[M_m]$ and $\mathbb{E}[J_i J_j]$ for $i \ne j$. Deduce

$$\mathrm{Var}(M_m) = nq(1 - q) + n(n - 1)(b - q^2).$$

Are missing-type indicators independent? (4)

(b) Derive the exact completion CDF by inclusion–exclusion: $\mathbb{P}(T_n \le m) = \sum_{j=0}^{n} (-1)^j \binom{n}{j} (1 - j/n)^m$. (3)

(c) For six types, compute $\mathbb{P}(T_6 \le 15)$ and $\mathbb{E}[M_{15}]$ to four decimals. Find the smallest integer m with $\mathbb{P}(T_6 \le m) \ge .95$; report probabilities at $m - 1$ and m. (4)

(d) Prove $\mathbb{P}(T_n > m) \le n e^{-m/n}$. For 100 types choose an integer m guaranteeing at least 95% completion by this bound. Does the bound establish the smallest possible deadline? (3)

III. Densities and tail areas

5. A density that exceeds one [12 points]

Suppose $f_X(x) = cx^2$ for $0 \le x \le 2$, and zero elsewhere.

(a) Normalize the density and write its full piecewise CDF. (3)

(b) Find $\mathbb{P}(1/2 < X \le 1)$, $\mathbb{E}[X]$, and $\mathrm{Var}(X)$. (3)

(c) Find the median and $\mathbb{P}(X = 1)$. Explain why $f_X(2) > 1$ is allowed. (3)

(d) Let $Y = (X/2)^3$. Find $F_Y(y)$ by rewriting $\{Y \le y\}$ as an event about X, and identify the distribution of Y. (3)

6. Tail areas and the suspiciously generous prize [12 points]

For a real variable Z, write $F(z) = \mathbb{P}(Z \le z)$ and assume $\mathbb{E}[|Z|] < \infty$ in (a)–(b).

(a) For $Z_+ = \max(Z, 0)$ and $Z_- = \max(-Z, 0)$, use $y = \int_0^\infty \mathbf{1}_{\{y>t\}}\, dt$ for $y \ge 0$ to show

$$\mathbb{E}[Z] = \int_0^\infty (1 - F(t))\, dt - \int_{-\infty}^0 F(t)\, dt.$$

You may interchange expectation and nonnegative integrals. Explain why using $\mathbb{P}(Z < t)$ instead of $\mathbb{P}(Z \le t)$ does not change an integral. (3)

(b) What does the *sum* of these two nonnegative tail areas compute? Check both identities for the constant payoff $Z = -2$. (3)

(c) A carnival advertises a prize R (in dollars) with CDF zero for $x < 1$ and $F_R(x) = 1 - 1/x$ for $x \ge 1$. Show that this is a valid continuous law, find its density, and use a tail integral to find its mean. Is there a finite fair entry fee for the uncapped game? (3)

(d) The small print caps the payout at $B > 1$ dollars: $C = \min(R, B)$. Use its tail to find the fair entry fee. What probability mass is at B? Evaluate the fee for $B = 100$. (3)

IV. Continuous mechanisms and a signed payoff

7. The auction for the last slice [12 points]

At a pizza party, $n \geq 1$ hungry mathematicians independently submit bids $U_1, \ldots, U_n$, each uniform on $[0, 1]$ dollars. The seller receives the *highest bid* $M = \max_i U_i$, even if it is small. The slice cost c dollars, where $0 < c < 1$, so the seller's net payoff is $Z = M - c$.

(a) Find the full CDF of M, and then of Z, using independence. (3)

(b) Compute separately the negative-tail area $L = \int_{-\infty}^{0} F_Z(z)\,dz$ and positive-tail area $G = \int_{0}^{\infty}(1 - F_Z(z))\,dz$. Use $G - L$ to find the mean profit *without differentiating the CDF*. (4)

(c) Find the loss probability and $\mathbb{E}[|Z|]$. Explain why the latter need not equal the absolute value of the mean profit. (3)

(d) For $n = 3, c = 3/4$, calculate the mean profit, loss probability, and mean absolute payoff. What does this say about calling a game "fair"? (2)

8. A clock, a bus, and a normal measurement [14 points]

(a) A service has independent Poisson counts on disjoint intervals, with mean $3t$ requests in t minutes. Let W be the time to the next request. Derive its survival function from a zero-count probability; find its mean, variance, and median. (4)

(b) Compute $\mathbb{P}(W > 5/3 \mid W > 1)$ and compare it with $\mathbb{P}(W > 2/3)$. A bus wait V is instead uniform on $[0, 2]$ minutes. Compute $\mathbb{P}(V > 5/3 \mid V > 1)$ and $\mathbb{P}(V > 2/3)$; compare. (4)

(c) The duration X of a task is modeled as normal with mean 10 and variance 9 minutes squared. Express $\mathbb{P}(8 < X < 13)$ and $\mathbb{P}(X > 13)$ in terms of the standard normal CDF Φ. (3)

(d) What density-height or endpoint mistake appears in "$\mathbb{P}(X = 10) = 1/(3\sqrt{2\pi})$"? For $Y = 2X - 5$, give the distribution, mean, and variance. (3)

Optional challenge: the rare sticker [10 extra points]

Now type i has probability $p_i > 0$, with $\sum_{i=1}^{n} p_i = 1$. Purchases are still independent. Let T count draws until all types appear. Write $p_A = \sum_{i \in A} p_i$.

(a) Explain why the next-new-type probability after k types have appeared can depend on *which* types they were. Derive $\mathbb{P}(T > m) = \sum_{\varnothing \neq A \subseteq \{1,\ldots,n\}} (-1)^{|A|+1}(1 - p_A)^m$ for $m \geq 1$. (3)

(b) Use the discrete tail sum to obtain $\mathbb{E}[T] = \sum_{\varnothing \neq A} (-1)^{|A|+1}/p_A$. For three types with probabilities $1/2, 1/3, 1/6$, evaluate it and compare with three equally likely types. (4)

(c) For two types with probabilities $p, 1 - p$, show the mean is $1/p + 1/(1 - p) - 1$ and is minimized uniquely at $p = 1/2$. (3)